Question 4: Neural Network Derivation

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Problem Statement

Consider the neural network shown in Figure 5 for a binary classification task. Assume the hidden layer uses a linear activation function:

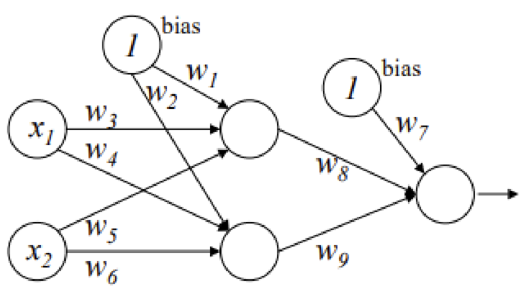
$$h(z) = cz$$

and the output layer uses the sigmoid function:

$$g(z) = \frac{1}{1 + e^{-z}}$$

The network wants to learn a function for:

$$P(Y = 1 \mid X, W) \quad \text{where} \quad W = (w_1, w_2, \ldots, w_9), \quad X = (x_1, x_2)$$



شکل ۵: شبکهٔ عصبی سوال چهارم.

Figure 5: The neural network architecture for Question 4

1. Derive the final output of the network $P(Y = 1 \mid X, W)$

First, compute the activations at the hidden layer:

$$Z'_1 = w_3 x_1 + w_5 x_2 + w_1$$
$$Z'_2 = w_4 x_1 + w_6 x_2 + w_2$$

Then compute the output before the activation function:

$$y' = (cw_8w_3 + cw_9w_4)x_1 + (cw_8w_5 + cw_9w_6)x_2 + cw_1 + cw_2 + w_7$$

Final output:

$$y = g(y') = \frac{1}{1 + e^{-y'}}$$

2. Write the decision boundary

On the decision boundary:

$$y = \frac{1}{1 + e^{-y'}} = 0.5 \Rightarrow y' = 0$$

Thus:

$$a_1x_1 + a_2x_2 + b = 0$$

where:

$$\begin{aligned} a_1 &= cw_8w_3 + cw_9w_4 \\ a_2 &= cw_8w_5 + cw_9w_6 \\ b &= cw_1 + cw_2 + w_7 \end{aligned}$$

3. Can this be done without a hidden layer?

Yes, if a direct linear combination of x_1 and x_2 using appropriate weights could approximate the same decision boundary, it is possible to eliminate the hidden layer. However, the hidden layer provides more flexibility in representing complex boundaries.